

//Exercise 3: Stored Procedures

Scenario 1: The bank needs to process monthly interest for all savings accounts.

Question: Write a stored procedure ProcessMonthlyInterest that calculates and updates the balance of all savings accounts by applying an interest rate of 1% to the current balance.

```
CREATE OR REPLACE PROCEDURE ProcessMonthlyInterest
```

```
IS
```

```
BEGIN
```

```
    UPDATE SavingsAccounts
```

```
    SET Balance = Balance + (Balance * 0.01);
```

```
    COMMIT;
```

```
END;
```

```
BEGIN
```

```
    ProcessMonthlyInterest;
```

```
END;
```

Query result	Script output	DBMS output	Explain Plan	SQL history
  Download ▾ Execution time: 0.007 seconds				
	ACCOUNTID	CUSTOMERNAME	BALANCE	
1	101	John	10100	
2	102	Alice	15150	
3	103	Bob	20200	

Scenario 2: The bank wants to implement a bonus scheme for employees based on their performance.

Question: Write a stored procedure **UpdateEmployeeBonus** that updates the salary of employees in a given department by adding a bonus percentage passed as a parameter.

```
CREATE OR REPLACE PROCEDURE UpdateEmployeeBonus(
```

```
    p_department IN VARCHAR2,
```

```
    p_bonus_percent IN NUMBER
```

```
)
```

```
IS
```

```
BEGIN
```

```
    UPDATE Employees
```

```
    SET Salary = Salary + (Salary * p_bonus_percent / 100)
```

```
    WHERE Department = p_department;
```

```
    COMMIT;
```

```
END;
```

```
BEGIN
```

```
    UpdateEmployeeBonus('IT', 10);
```

```
END;
```

Query result					Script output	DBMS output	Explain Plan	SQL history
  Download ▾ Execution time: 0.009 seconds								
	EMPLOYEEID	EMPLOYEENAME		DEPARTMENT	SALARY			
1	1	Raj		IT	55000			
2	2	Priya		IT	66000			
3	3	Arun		HR	45000			

Scenario 3: Customers should be able to transfer funds between their accounts.

Question: Write a stored procedure **TransferFunds** that transfers a specified amount from one account to another, checking that the source account has sufficient balance before making the transfer.

```
CREATE OR REPLACE PROCEDURE TransferFunds(
```

```
    p_from_account IN NUMBER,
```

```
    p_to_account IN NUMBER,
```

```
    p_amount IN NUMBER
```

```
)
```

```
IS
```

```
    v_balance NUMBER;
```

```
BEGIN
```

```
    SELECT Balance
```

```
    INTO v_balance
```

```
    FROM Accounts
```

```
    WHERE AccountID = p_from_account;
```

```
    IF v_balance >= p_amount THEN
```

```
        UPDATE Accounts
```

```
        SET Balance = Balance - p_amount
```

```
        WHERE AccountID = p_from_account;
```

```
        UPDATE Accounts
```

```
        SET Balance = Balance + p_amount
```

```
        WHERE AccountID = p_to_account;
```

```
    COMMIT;
```

```
ELSE  
    DBMS_OUTPUT.PUT_LINE('Insufficient Balance');  
END IF;  
END;  
  
BEGIN  
    TransferFunds(1, 2, 3000);  
END;
```

Query result					Script output					DBMS output					Explain Plan					SQL history				
										Download					Execution time: 0.008 seconds									
	ACCOUNTID				CUSTOMERNAME				BALANCE															
1	1				John				7000															
2	2				Alice				8000															